

EXPLAINABLE AI-BASED FINGERPRINT SPOOF DETECTION SYSTEM FOR DIGITAL FORENSICS

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by

K. M. Pasindu Chanuka Elapatha

Supervised by

Ms. Hirushi Dilpriya

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Chapter 1 - Introduction

1.1 Chapter Overview

This chapter introduces the research undertaken in the field of explainable artificial intelligence (XAI) applied to fingerprint presentation attack detection (PAD) for digital forensic use. Section 1.2 establishes the background by describing the ubiquity of fingerprint biometrics, the escalating threat posed by presentation attacks fabricated from materials such as silicone and gelatin, and the academic lineage of the Liveness Detection (LivDet) competitions. Section 1.3 articulates the general and specific problem, leading into the explicit research gap. Section 1.4 frames the research question, while Sections 1.5 and 1.6 set out the motivation and aim. Section 1.7 enumerates the four research objectives that operationalise the aim. Section 1.8 presents a rich picture of the proposed solution, Section 1.9 lists the hardware and software resource requirements, Section 1.10 demarcates the project scope, and Section 1.11 closes with a summary that signposts the literature review undertaken in Chapter 2.

1.2 Problem Background

Fingerprint biometric authentication has become one of the most pervasive identity verification technologies of the twenty-first century. India's Aadhaar programme alone has enrolled in excess of 1.3 billion residents, indexing each with a tenprint capture, while consumer-grade fingerprint sensors are embedded in smartphones, laptops, banking kiosks, attendance terminals, border-control e-gates and forensic livescan stations. The appeal of the modality is well documented: ridge patterns are believed to be distinctive, persistent across the human life-span, and capturable with low-cost optical or capacitive hardware [1]. Consequently, organisations have increasingly substituted password-based or token-based authentication with fingerprint matching, on the assumption that the physical presence of an authorised finger is a prerequisite to a positive match.

That assumption has, however, been progressively eroded by the maturation of presentation attack techniques. A presentation attack, as standardised by ISO/IEC 30107, is any deliberate attempt to subvert a biometric capture subsystem using an artefact

placed at the sensor. In the fingerprint domain, common artefacts include moulds cast from silicone, gelatin, latex, wood glue, Ecoflex, Play-Doh and Body Double, often produced from latent prints lifted from glass or plastic surfaces. Academic literature has demonstrated that even modestly resourced adversaries can fabricate spoofs that defeat commodity sensors whose decision logic is restricted to template matching [2]. The phenomenon is no longer confined to the laboratory: it has migrated into operational fraud.

Recent incidents make the operational threat concrete. In January 2025, four government employees in Kuwait were arrested for using silicone replicas of colleagues' fingertips to register false attendance on biometric clocking devices, enabling absentee fraud over an extended period. In a separate Kuwaiti investigation conducted by the General Department of Criminal Evidence, seven court secretaries were found to have employed wax-based fingerprint spoofs to authenticate the falsification of judicial records, an offence with direct implications for evidentiary integrity. Comparable cases have been documented in attendance fraud in South Asia, in welfare-benefit collection in West Africa, and in unauthorised access to safe-deposit facilities in Europe. These incidents demonstrate that fingerprint spoofing has matured from an academic curiosity into a vector for criminal and administrative fraud at the institutional scale.

In parallel with the rise of the threat, the academic community has constructed a standardised evaluation infrastructure. The Liveness Detection (LivDet) competition series, inaugurated in 2009 and recurring biennially, releases curated datasets containing both live (*bona fide*) and spoof captures collected across multiple sensors and fabrication materials. Successive editions in 2009, 2011, 2013, 2015, 2017, 2019, 2021 and 2023 have widened the diversity of sensors and artefact materials and have established the Average Classification Error (ACE), Attack Presentation Classification Error Rate (APCER) and Bona fide Presentation Classification Error Rate (BPCER) as the *de facto* reporting metrics. The competition has driven the field from handcrafted texture descriptors toward deep convolutional architectures.

Modern convolutional neural network (CNN) classifiers routinely report accuracies in excess of 97% on within-dataset LivDet evaluations [6]. Hybrid CNN architecture

combining VGG16 and ResNet50 have reported 99.72% on LivDet 2013, and lightweight residual designs such as Slim-ResCNN secured first place in LivDet 2017 [5]. Yet, two systemic deficiencies persist in this otherwise impressive literature. Firstly, the decision logic of these CNNs remains opaque: the models output a liveness score without articulating which ridge or pore features informed the verdict. Secondly, performance degrades substantially when the model is evaluated on sensors or materials not represented in training, an issue formalised as intra-dataset overfitting or cross-sensor brittleness [3][5]. The intersection of these two deficiencies frames the research undertaken in this dissertation.

1.3 Problem Statement

1.3.1 General Problem

The general problem addressed by this research is that contemporary CNN-based fingerprint presentation attack detection systems, although accurate within their training distribution, operate as black boxes whose decisions cannot be inspected, audited or justified to a non-technical stakeholder. In a forensic deployment, the consumer of the model output is not a machine-learning engineer but a fingerprint examiner, an investigator, a prosecutor, a defence counsel or a trier of fact. Each of these stakeholders requires not merely a binary verdict of "live" or "spoof" but a transparent rationale that can be entered into the chain of evidentiary reasoning. Without such transparency, the model output occupies an uncomfortable epistemological position: it is too consequential to ignore yet too opaque to credit.

This opacity is incompatible with the prevailing legal frameworks governing the admissibility of scientific evidence. The Daubert standard in the United States, the Frye general-acceptance standard in some state jurisdictions, and analogous gatekeeping doctrines in the United Kingdom, the European Union and the Commonwealth all demand that expert evidence be founded on a method that is testable, has been subjected to peer review, has a known or potential error rate, and is generally accepted within the relevant scientific community. A model that cannot explain its decision cannot, in any meaningful sense, be tested for its reasoning, nor can the error rate of that reasoning be

characterized. The general problem is therefore that high-accuracy black-box PAD systems remain forensically inadmissible despite their technical maturity.

1.3.2 Specific Problem

Within the fingerprint PAD literature, the specific manifestation of this problem has three intertwined dimensions. The first is intra-dataset overfitting: models that achieve over 95% accuracy on a single LivDet edition routinely collapse when evaluated on a different edition, a different sensor, or a different fabrication material, with reported degradations of ten to twenty percentage points [5]. The second is cross-sensor brittleness: domain shifts in capture device, resolution and pre-processing pipeline produce systematic distributional differences that are invisible at the pixel level yet decisive at the feature level, undermining generalization [3]. These twin deficiencies indicate that headline accuracy figures are not a reliable indicator of operational performance.

The third and least addressed dimension is the absence of practitioner-validated explanations. Although general-purpose explainable artificial intelligence methods such as Grad-CAM++ [12], SHAP [13] and LIME [14] have been applied to medical imaging, autonomous driving and natural language processing, their application to fingerprint PAD has been sporadic, single-method, and rarely subjected to quantitative faithfulness evaluation. Critically, no published study has triangulated multiple XAI methods on the same fingerprint PAD model, measured their cross-method agreement, and elicited the perceptions of working forensic practitioners on the trustworthiness of the resulting visualisations. The fingerprint PAD literature therefore lacks both a quantitatively grounded multi-method XAI comparison and a practitioner-centred study of acceptability.

The specific research gap addressed by this dissertation is therefore the absence of a forensic-grade fingerprint PAD pipeline that simultaneously (a) supports multiple, theoretically distinct XAI methods on the same underlying CNN backbone, (b) quantifies the faithfulness of each XAI method using established deletion, insertion and cross-method agreement protocols [15][20], and (c) is presented to practitioners through a workflow aligned with Daubert and Frye reporting expectations. Closing this gap requires

not a new CNN architecture in isolation but the integration of architecture, explanation, evaluation and forensic workflow into a single coherent artefact.

1.4 Research Question

The principal research question that frames this dissertation is the following: How can explainable artificial intelligence techniques be integrated with convolutional neural network fingerprint presentation attack detection to produce decisions that are simultaneously accurate, faithfully explained, and acceptable to forensic practitioners under Daubert and Frye admissibility standards?

From this principal question, four subsidiary questions follow:

1. Which CNN backbone architecture among ResNet50V2, ResNet50 with Convolutional Block Attention Module (CBAM) and MobileNetV3-Large offers the most favourable trade-off between classification error, inference latency, parameter count and explanation generation cost on the combined LivDet 2009–2015 corpus?
2. To what extent do Grad-CAM++, SHAP DeepExplainer and LIME produce explanations that are mutually consistent, and how is their faithfulness quantified by deletion-AUC, insertion-AUC, Pearson correlation and Intersection-over-Union at the top-twenty-per-cent salient region?
3. How can a forensic case-management workflow including audit trail, case metadata strip and PDF reporting be constructed so as to render the model's decision and its explanation admissible under the Daubert and Frye gatekeeping standards?
4. How do fingerprint examiners, biometric researchers and digital forensic investigators perceive the trustworthiness, interpretability and operational utility of the resulting XAI-augmented PAD system when surveyed through a structured 32-item instrument?

1.5 Research Motivation

The motivation for this research is shaped by three converging trajectories. The first is the documented escalation of presentation-attack-based fraud in operational settings. The arrest of four Kuwaiti government employees in January 2025 for silicone-based

attendance fraud, the parallel investigation into seven court secretaries using wax spoofs to falsify judicial records, and the proliferation of low-cost spoof-fabrication tutorials on open online platforms collectively demonstrate that the threat is no longer confined to academic adversarial models. Each documented incident reinforces the urgency of deploying PAD that is not only accurate but defensible when the resulting evidence is contested in administrative or criminal proceedings.

The second motivating trajectory is the cautious yet decisive adoption of artificial intelligence within forensic science. National forensic institutes in the United Kingdom, the United States, Germany and the Netherlands have published statements indicating that AI-assisted analysis is permissible only where the underlying model is interpretable and where the error characteristics are documented. The European Union Artificial Intelligence Act, enacted in 2024 and progressively coming into force, classifies biometric identification used by law-enforcement authorities as high-risk and mandates transparency, human oversight and technical documentation. This regulatory environment renders black-box CNN PAD untenable for deployment, regardless of its headline accuracy.

The third trajectory is the persistent gap in practitioner-centred research. The vast majority of published fingerprint PAD studies report only quantitative metrics on benchmark partitions and do not engage the human experts whose work the technology purports to support. There is, accordingly, scant empirical evidence on whether forensic examiners find XAI visualisations interpretable, whether they trust them, and whether they would integrate them into casework. Daubert and Frye both incorporate a notion of acceptance by the relevant expert community, which cannot be inferred from leaderboard performance alone. This dissertation therefore seeks to combine quantitative model evaluation with qualitative practitioner consultation, addressing both the technical and the socio-legal dimensions of admissibility.

1.6 Research Aim

The aim of this research is to design, implement and evaluate an explainable artificial intelligence-based fingerprint spoof detection system that produces accurate live-versus-

spoof classifications using a convolutional neural network backbone, accompanies each classification with theoretically distinct explanations generated by Grad-CAM++, SHAP and LIME, quantifies the faithfulness of those explanations through deletion, insertion and cross-method agreement protocols, exposes the resulting decisions through a forensic workflow comprising case management, audit trail and Daubert/Frye-aligned PDF reporting, and validates the acceptability of the system through a structured survey of fingerprint examiners, biometric researchers and digital forensic investigators, thereby producing a candidate artefact for operational use in digital forensic investigations.

1.7 Research Objectives

1.7.1 Objective 1

To identify, through a systematic review of the recent fingerprint presentation attack detection and explainable artificial intelligence literature, the dominant CNN architectures, XAI techniques, faithfulness evaluation protocols and admissibility frameworks relevant to forensic deployment, and to specify the resulting research gap.

1.7.2 Objective 2

To analyse the combined LivDet 2009, 2011, 2013 and 2015 corpus comprising approximately sixty-five thousand fingerprint images across nine sensors and twelve fabrication materials, characterising class balance, sensor distribution and material distribution, and to derive a stratified train/test partition that preserves operational diversity.

1.7.3 Objective 3

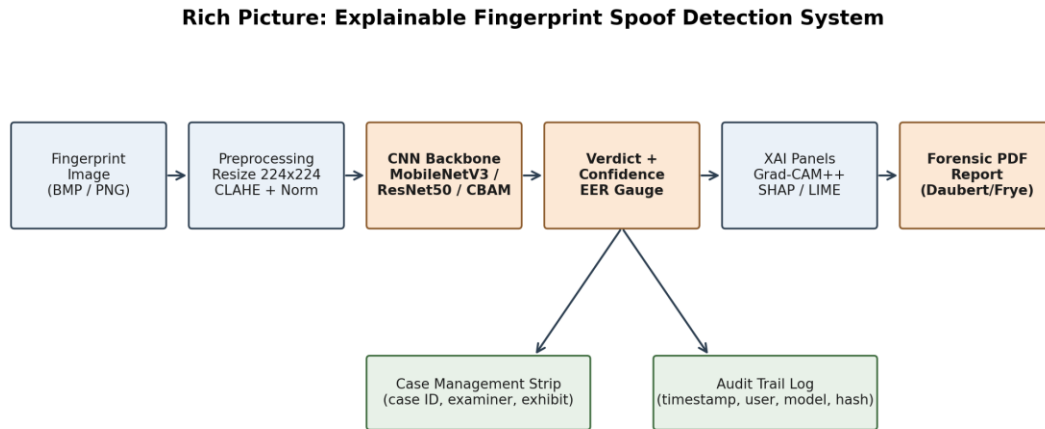
To design and implement a fingerprint PAD pipeline that trains three CNN backbones (ResNet50V2, ResNet50 with CBAM and MobileNetV3-Large), integrates Grad-CAM++, SHAP DeepExplainer and LIME explanation generators, and exposes the resulting decisions through a Streamlit-based forensic application with case management, audit trail and fpdf2-based PDF reporting aligned to Daubert and Frye expectations.

1.7.4 Objective 4

To evaluate the implemented system quantitatively using AUC, ACE, APCER, BPCER, deletion-AUC, insertion-AUC, cross-method Pearson correlation and IoU at the top-twenty-per-cent salient region, and qualitatively through a 32-item, nine-section practitioner survey targeting fingerprint examiners, biometric researchers and digital forensic investigators.

1.8 Rich Picture of the Proposed Solution

The rich picture in Figure 1.1 depicts the end-to-end workflow of the proposed system, situating each component within its operational context. The diagram is organised around four conceptual zones: the data zone on the left, where LivDet images and case evidence enter the pipeline; the modelling zone in the centre, where the CNN backbones and XAI generators reside; the forensic-application zone on the right, where the Streamlit interface, audit trail and PDF reports interact with the human user; and the stakeholder zone surrounding the system, identifying the forensic examiner, the biometric researcher, the supervising investigator and the legal admissibility framework as primary actors.



End-to-end forensic pipeline: image acquisition through explainable verdict to court-admissible report.

Figure 1.1: Rich Picture of the Proposed Solution

The workflow proceeds as follows. A fingerprint image, whether drawn from the LivDet benchmark for evaluation or captured at a forensic livenesscan station for casework, is

uploaded to the Streamlit application together with case metadata such as case identifier, examiner name and chain-of-custody timestamp. The image is pre-processed (resized to 224 by 224 pixels, normalised against ImageNet statistics) and passed to the selected CNN backbone, which yields a liveness probability. The selected XAI generators are then invoked: Grad-CAM++ produces a gradient-weighted class-activation map, SHAP DeepExplainer produces Shapley value attributions at the pixel level, and LIME perturbs superpixels to estimate a locally linear surrogate model.

The three explanations are rendered side-by-side together with the quantitative faithfulness measures (deletion-AUC, insertion-AUC and inter-method agreement). All actions are recorded in an immutable audit trail, and the final report is rendered to a Daubert/Frye-aligned PDF that contains the case metadata, the decision, the three explanations, the faithfulness metrics and a methodological declaration. The report is the artefact that ultimately enters the evidentiary process, and it is the design of this artefact, rather than the CNN in isolation, that closes the loop between machine learning and admissible forensic practice.

1.9 Resource Requirements

1.9.1 Hardware

The hardware required for the development, training, evaluation and demonstration of the proposed system is summarised in Table 1.1. The development workstation supports day-to-day coding, dataset curation, application development and CPU-side inference, while training of the deeper backbones is offloaded to a cloud GPU environment to keep electricity and thermal load within domestic limits.

Component	Specification	Purpose
Development workstation	Windows 11 Pro, Intel Core i7 (8 cores), 16 GB RAM	Coding, dataset curation, Streamlit UI development, CPU-side inference and XAI on smaller models
GPU resource	Google Colab T4 (16 GB VRAM) free tier with Pro	Training of ResNet50V2, ResNet50+CBAM and

	fallback	MobileNetV3-Large; SHAP DeepExplainer batches
Local storage	500 GB SSD with at least 50 GB free	Storage of LivDet 2009–2015 corpus (~12 GB), checkpoints, XAI artefacts and PDF reports
Display	Dual-monitor 1080p setup	Concurrent code, Streamlit preview and reference reading during development
Network	Stable broadband (minimum 20 Mbps)	Dataset download from livdet.org, Colab session synchronisation, GitHub push/pull

Table 1.1: Hardware Resource Requirements

1.9.2 Software

The software stack, summarised in Table 1.2, is centred on the Python scientific computing ecosystem. PyTorch 2.x with torchvision supplies the ImageNet pretrained backbones; pytorch-grad-cam, shap and lime supply the XAI methods; Streamlit provides the forensic interface; fpdf2 and Jinja2 generate the PDF reports; and Git with GitHub provides version control and reproducibility.

Category	Tool	Purpose
Language and runtime	Python 3.13	Primary implementation language for all modules
Deep learning	PyTorch 2.x, torchvision	CNN backbone definition, training loop, ImageNet pretrained weights
Scientific computing	NumPy, pandas, scikit-learn, matplotlib, seaborn	Data manipulation, metric computation, plotting of results
Explainable AI	pytorch-grad-cam, shap (GradientExplainer fallback), lime	Generation of Grad-CAM++, SHAP and LIME explanations

Forensic application	Streamlit	Web-based UI for case management, upload, analysis and reporting
Reporting	fpdf2, Jinja2	Generation of Daubert/Frye-aligned PDF case reports
Version control	Git, GitHub	Source code management, supervisor review, reproducibility
Development environment	VS Code	Code authoring, debugging and GPU-backed training
Documentation	Microsoft Word, Mendeley	Dissertation drafting and citation management

Table 1.2: Software Resource Requirements

1.10 Project Scope

Table 1.3 demarcates the items that fall within the scope of this dissertation from those that lie outside it. The scope is delimited in order to make the work tractable within the November 2026 submission window while still producing a coherent forensic-grade artefact.

In Scope	Out of Scope
Fingerprint presentation attack detection on the combined LivDet 2009–2015 corpus	Face, iris, voice or multimodal biometric presentation attack detection
Three CNN backbones: ResNet50V2, ResNet50+CBAM and MobileNetV3-Large	Vision Transformer, Swin or diffusion-based PAD architectures
Three XAI methods: Grad-CAM++, SHAP DeepExplainer and LIME	Counterfactual explanations, concept activation vectors or natural-language rationales
Quantitative XAI faithfulness via deletion-AUC, insertion-AUC, Pearson correlation and IoU	Adversarial robustness evaluation under PGD, FGSM or AutoAttack threat models
Streamlit forensic application with case management, audit trail and PDF reporting	Mobile-native (Android/iOS) deployment or embedded firmware integration

Practitioner survey of fingerprint examiners, biometric researchers and digital forensic investigators	Live deployment within an operational forensic laboratory or police authority
Reporting aligned to Daubert and Frye admissibility standards	Formal legal certification, ISO/IEC 17025 accreditation or court-ready validation
Open, publicly available LivDet datasets only	Proprietary, classified or operational fingerprint case data
Liveness/spoof binary classification	Identification (1:N), verification (1:1) or latent print matching
Windows 11 desktop development and Google Colab T4 training	Custom hardware acceleration, FPGA or embedded sensor co-design

Table 1.3: Project Scope (Inclusions and Exclusions)

The rationale for these scope decisions is threefold. Firstly, restricting the biometric modality to fingerprints and the task to binary liveness classification preserves analytical depth, given the breadth already introduced by three backbones and three XAI methods. Secondly, restricting the datasets to LivDet 2009-2015 ensures full reproducibility for external reviewers and avoids the legal and ethical complications associated with proprietary or operational data. Thirdly, framing the survey as a practitioner perception study rather than as an operational pilot keeps the empirical work within the ethical clearance footprint achievable for an undergraduate dissertation while still producing genuinely novel evidence on stakeholder acceptability.

1.11 Chapter Summary

This chapter has established the rationale for an explainable artificial intelligence-based fingerprint spoof detection system designed for digital forensic use. It has documented the real-world escalation of presentation attack fraud, illustrated by the 2025 Kuwaiti silicone attendance and wax judicial fraud cases, and has shown that, while CNN-based PAD has matured to exceed 97% accuracy on LivDet benchmarks, its black-box character is incompatible with the Daubert and Frye admissibility standards. The research gap was characterised as the absence of a multi-method, faithfulness-evaluated, practitioner-validated XAI pipeline for fingerprint PAD; the principal research question

and four subsidiary questions were articulated; the aim and four research objectives were stated; the rich picture and resource requirements were presented; and the project scope was demarcated.

Chapter 2 builds on this foundation by undertaking a structured literature review covering, in turn, fingerprint presentation attack detection techniques from handcrafted descriptors to modern deep architectures, attention mechanisms and lightweight CNN designs, the XAI methods Grad-CAM++, SHAP and LIME, the faithfulness evaluation protocols of deletion and insertion AUC, the legal admissibility frameworks of Daubert and Frye, and the methodological positioning of Design Science Research. The review culminates in a synthesised research gap that justifies, in finer empirical detail, the artefact developed and evaluated in the subsequent chapters.

Chapter 2 - Literature Review

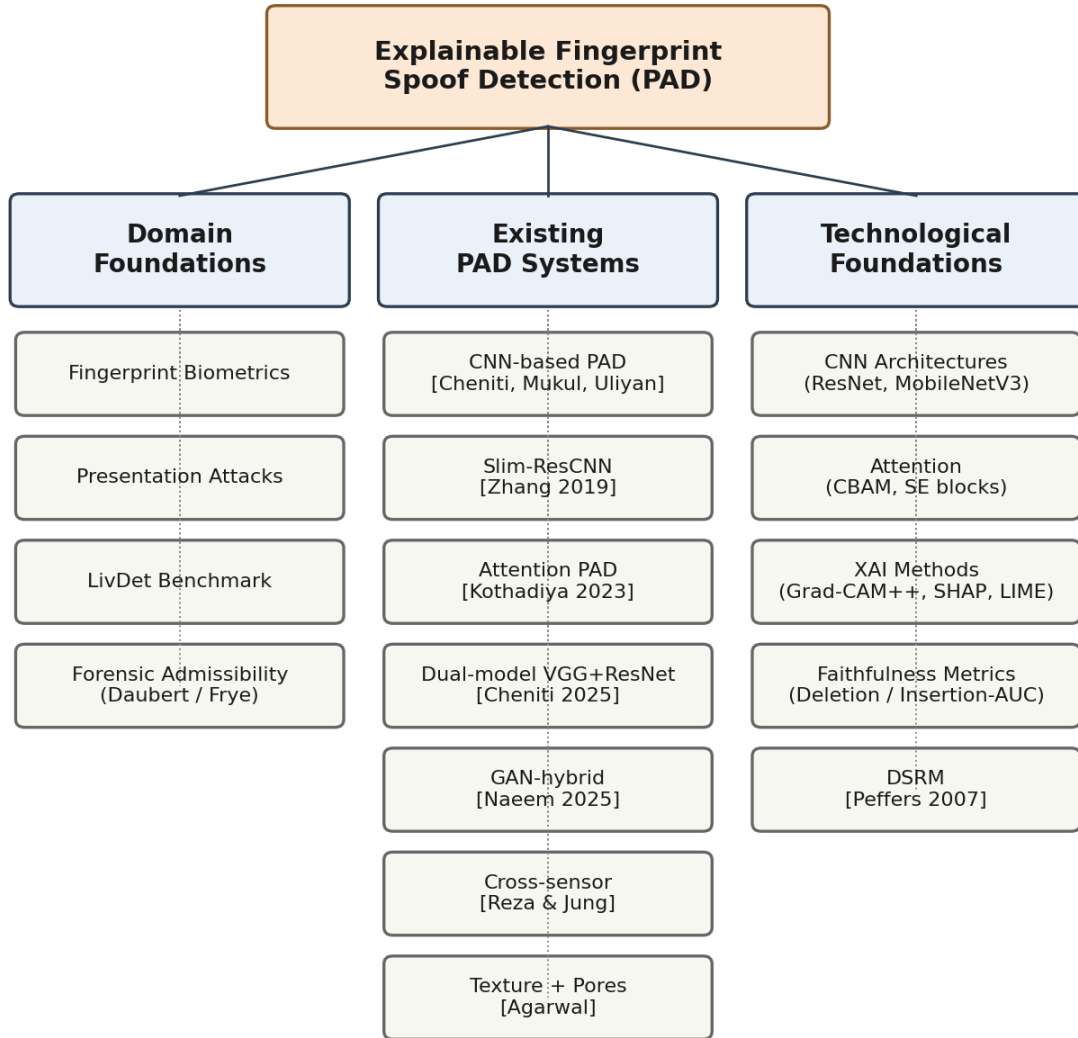
2.1 Chapter Overview

This chapter examines the body of academic and industrial literature surrounding fingerprint presentation attack detection (PAD), with a particular focus on convolutional neural network (CNN) architectures, attention mechanisms, explainable artificial intelligence (XAI), and the forensic admissibility frameworks that govern the deployment of biometric evidence. The review is organised in accordance with the NSBM Green University Faculty of Computing literature review structure, allocating approximately 10% of the discussion to the domain overview, 30% to existing systems and frameworks, and 60% to a technological analysis covering algorithmic, design, and workflow dimensions. The chapter concludes with a reflective synthesis that articulates the research gap addressed by the present project and outlines how the subsequent chapter operationalises the identified shortcomings into a coherent methodology.

2.2 Conceptual Map of the Literature

The conceptual map presented in Figure 2.1 organises the surveyed literature along three interlocking axes. The horizontal axis traces the historical evolution of fingerprint PAD from handcrafted texture descriptors through early CNN-based detectors to attention-enhanced and generative-adversarial hybrid architectures. The vertical axis distinguishes the model layer (backbones, attention modules, loss formulations) from the explanation layer (gradient-based, perturbation-based, and Shapley-value-based attribution methods) and from the deployment layer (forensic admissibility, audit logging, practitioner workflow). A third dimension overlays evaluation paradigms, separating intra-dataset accuracy reporting from cross-sensor and cross-material generalisation studies. This conceptual organisation allows the reader to locate each cited work within a multi-dimensional space and exposes the comparatively sparse intersection of attention-enhanced PAD, multi-method XAI faithfulness evaluation, and practitioner-informed forensic deployment, which is the empty cell that this project seeks to populate.

Conceptual Map of the Literature



Synthesis of domain knowledge, prior PAD work, and enabling technologies underpinning this study.

Figure 2.1: Conceptual Map of the Literature

2.3 Domain Overview

Fingerprint biometrics has emerged over the past four decades as the dominant modality for large-scale civil and forensic identification, largely on the strength of its favourable

balance between distinctiveness, permanence, and acquisition cost. National identity programmes such as India's Aadhaar have enrolled in excess of 1.3 billion individuals through ten-print and dual-iris capture, while law enforcement agencies worldwide rely on Automated Fingerprint Identification Systems (AFIS) to search latent prints against multi-million record databases. Beyond government use, fingerprint sensors are now embedded in consumer smartphones, banking terminals, attendance systems, and access-control infrastructure, exposing the modality to operational environments of widely varying integrity assurance. The ubiquity of the technology, however, has rendered it an attractive target for adversarial circumvention, motivating sustained research into presentation attack detection as a complement to template matching [2].

A presentation attack, as defined by ISO/IEC 30107-1, is the deliberate presentation of an artefact or altered biometric trait to the sensor with the intention of interfering with the normal operation of the recognition system. In the fingerprint domain, attack artefacts are typically fabricated from silicone, gelatin, latex, wood glue, Play-Doh, Ecoflex, Body Double, or related elastomers and colloids cast from a cooperative or covertly acquired mould of a genuine finger [1][5]. The fabrication process exploits the optical and capacitive response of common sensors to subcutaneous moisture, ridge geometry, and conductivity, and the resulting artefacts can defeat sensors that rely solely on minutiae extraction and template matching. Recent operational incidents illustrate the urgency of the problem: in January 2025, Kuwaiti authorities arrested four government employees for using silicone fingerprint replicas to defraud attendance systems, and a parallel case involving seven court secretaries who employed wax spoofs to falsify judicial records was investigated by the General Department of Criminal Evidence.

To support reproducible evaluation of countermeasures, the Liveness Detection (LivDet) competition has been organised biennially since 2009, releasing standardised datasets of live and spoof fingerprint images captured across a rotating panel of sensors and materials [5]. The successive editions have expanded the diversity of sensors from optical capacitive devices such as the Biometrika 400B and Identix DFR2100 to swipe and thermal sensors, and have introduced increasingly unseen spoof materials at test time to probe generalisation. Despite the maturity of the benchmark, the literature documents a persistent intra-dataset overfitting problem in which models achieving near-saturated

accuracy on the training distribution exhibit substantial degradation when evaluated across unseen sensors or materials [3][5]. This generalisation gap remains a central methodological concern for any work aspiring to forensic deployment.

The forensic deployment context imposes evidentiary constraints that exceed those of routine access control. Under the Daubert standard articulated by the United States Supreme Court in 1993 and the antecedent Frye general-acceptance standard, the admissibility of scientific evidence is contingent upon testability, peer review, known or knowable error rates, the existence of standards governing operation, and acceptance within the relevant scientific community. A black-box neural classifier that emits only a binary decision cannot readily satisfy these criteria, since neither its decision rationale nor its failure modes are inspectable by the trier of fact. This evidentiary lacuna provides the practical motivation for integrating explainable artificial intelligence into PAD pipelines intended for forensic use, and it frames the technological analysis that follows.

2.4 Existing Systems, Frameworks, and Designs

Early CNN-based PAD systems established the feasibility of replacing handcrafted texture descriptors with learned representations. Mukul and Lal [1] proposed a hybrid CNN architecture that fuses low-level texture statistics with mid-level convolutional features, reporting competitive accuracy on LivDet 2011 and 2013 subsets. While the hybrid formulation usefully demonstrates the complementarity of statistical and learned features, the authors do not report cross-sensor evaluation and the published implementation does not include any interpretability layer, leaving the model's decision rationale opaque to a forensic examiner. Uliyan et al. [8] extended the early CNN paradigm by introducing patch-based deep learning, in which overlapping image patches are independently classified and aggregated through a voting scheme. The patch-based design improves robustness to partial occlusions but inherits the same interpretability deficit and does not address the question of which patches contributed most to the final decision.

The Slim-ResCNN architecture proposed by Zhang et al. [5] represents a watershed contribution, as it won the LivDet 2017 competition with an overall accuracy of 95.25%

across the three released sensor subsets. The architecture introduces a slimmed residual block tailored to the relatively small spatial extent of fingerprint imagery and demonstrates that competition-grade performance is achievable without the parameter overhead of full ResNet50. Importantly, the same paper is among the first in the PAD literature to explicitly diagnose intra-dataset overfitting, observing that models trained on a single LivDet edition lose between five and twelve percentage points of accuracy when evaluated on a different edition. The authors propose data augmentation and dropout as partial remedies, but their evaluation does not extend to interpretability, and the model's attention to specific ridge or pore structures remains uncharacterised.

Attention-enhanced PAD architectures form the next significant strand of the literature. Kothadiya et al. [6] integrate a channel attention module into a ResNet50 backbone and report 97.78% accuracy and a bona fide presentation classification error rate (BPCER) of 0.89% on a curated LivDet subset. The work usefully demonstrates that the Convolutional Block Attention Module (CBAM) family of modules [10] can be inserted into a standard backbone with minimal architectural disruption while improving discrimination on subtle texture cues. However, the evaluation is restricted to a single sensor partition, and no analysis is offered of where the attention module concentrates its activations, which is precisely the question a forensic examiner would wish to ask. The absence of any explanation layer in an architecture explicitly built around attention is a notable omission that the present project addresses.

Cheniti, Akhtar, and Chandaliya [4] propose a dual-model fusion in which a VGG16 and a ResNet50 are independently fine-tuned and their logits combined through a learned weighting scheme, achieving 99.72% accuracy on LivDet 2013. The headline figure is impressive but is reported on the most homogeneous of the LivDet editions, and a supplementary cross-sensor evaluation included in the same paper reveals a drop of approximately twelve percentage points when the model is transferred to LivDet 2015 sensors. The authors candidly acknowledge this generalisation gap but do not propose a structural remedy, and they do not report any explainability analysis that might illuminate whether the two backbones attend to the same image regions or to complementary cues. The dual-model framework therefore offers a useful upper bound on intra-dataset

accuracy while reinforcing the need for cross-dataset combined training, which is the protocol adopted in the current work.

Naeem et al. [7] explore a generative-adversarial hybrid in which a generator synthesizes additional spoof samples to augment a CNN classifier, reporting 92.8% accuracy on a combined LivDet partition. The GAN-hybrid approach is conceptually attractive for addressing the class-imbalance and unseen-material problems, but the reported accuracy lags behind the attention-enhanced and dual-model baselines, and the synthetic samples themselves are not subjected to faithfulness analysis. Reza and Jung [3] focus exclusively on the cross-sensor generalization problem and propose a domain-adversarial training scheme that explicitly penalizes sensor-specific feature drift. Their work confirms that cross-sensor evaluation is the most punishing protocol in the PAD literature and that gains on this protocol are typically purchased at a modest cost in within-sensor accuracy, a trade-off that any forensic deployment must explicitly negotiate.

Agarwal and Bansal [9] depart from the pure-CNN paradigm by fusing pores, perspiration, and texture descriptors in a multi-stream classifier. The work usefully demonstrates that physiologically motivated features remain informative even in the era of deep learning, and the pore-channel attention activations are intuitively aligned with what a forensic examiner would inspect. The authors do not, however, quantify the alignment between their attention maps and any independent attribution method, and the multi-stream design imposes acquisition constraints (high-resolution capture for visible pores) that limit its applicability to legacy sensors. Taken together, the surveyed systems demonstrate that intra-dataset accuracy is approaching saturation, that attention mechanisms confer meaningful gains, and that the literature has consistently under-reported on interpretability, cross-dataset combined evaluation, and practitioner-facing deployment artefacts. These gaps frame the technological analysis that follows.

2.5 Technological Analysis

2.5.1 Algorithmic Analysis

The convolutional layer remains the fundamental building block of modern PAD architectures, mapping spatially correlated pixel neighbourhoods into hierarchical feature

representations through learnable filter banks. He et al. [17] introduced the residual learning framework in ResNet, in which a shortcut connection adds the identity of a block's input to its transformed output, thereby permitting the gradient to flow unimpeded through very deep networks. ResNet50, comprising fifty weighted layers organised into four bottleneck stages, has become a de facto baseline for fingerprint PAD on account of its strong ImageNet-pretrained representations and its relative parameter efficiency compared with VGG variants. In the present project, ResNet50V2 with pre-activation residual blocks served as one of the three evaluated backbones, achieving an area under the receiver operating characteristic curve (AUC) of 0.9925 and an average classification error (ACE) of 5.06% on the combined LivDet 2009-2015 test partition.

MobileNetV3-Large [11] represents a markedly different design philosophy. The architecture replaces standard convolutions with depthwise separable convolutions, in which spatial filtering and channel mixing are factorised into two sequential operations with substantially lower multiply-accumulate counts. Squeeze-and-excitation (SE) blocks recalibrate channel responses through a global pooling and gating mechanism, and the network's macro-architecture was discovered by neural architecture search rather than hand-designed. In the present work, MobileNetV3-Large emerged as the best overall model, recording an ACE of 4.60% and an AUC of 0.9924 while using approximately 82% fewer parameters than ResNet50 and delivering four-fold faster XAI generation on CPU. This result is significant because it inverts the common assumption that forensic-grade accuracy requires heavyweight backbones, and it materially lowers the barrier to on-premises deployment in laboratories without GPU infrastructure.

The Convolutional Block Attention Module (CBAM) introduced by Woo et al. [10] composes a channel attention branch with a spatial attention branch, each implemented through pooling, multilayer perceptron projection, and sigmoid gating. When inserted between the residual stages of a ResNet backbone, CBAM permits the network to suppress irrelevant feature channels and to emphasise informative spatial regions. The present project integrated CBAM after each of the four residual stages of ResNet50, producing the ResNet50+CBAM variant that achieved an AUC of 0.9901 and an ACE of 6.60%. The marginal accuracy degradation relative to the plain ResNet50 baseline is consistent with the observations of [6] and suggests that the benefit of attention in the

fingerprint domain is more pronounced on small or noisy datasets than on the relatively well-curated LivDet partitions used here. The attention maps themselves, however, provide an additional explanation channel that the plain backbones lack.

Loss formulation, optimisation, and learning-rate scheduling jointly determine the convergence behaviour of all three backbones. Binary cross-entropy was selected over focal loss given the natural fifty-one to forty-nine balance of the combined LivDet partition, which obviates the need for class weighting. The AdamW optimiser, which decouples weight decay from the adaptive moment estimates, was paired with cosine annealing of the learning rate from one times ten to the minus four to one times ten to the minus six over the training horizon. Explainability was operationalised through three complementary methods: Grad-CAM++ [12], which generalises gradient-weighted class activation mapping to handle multiple instances of the target class; SHAP DeepExplainer [13], which approximates Shapley values through DeepLIFT propagation rules; and LIME [14], which fits a sparse linear surrogate to local perturbations of superpixel masks. Faithfulness was quantified through the deletion-AUC and insertion-AUC protocols of Petsiuk et al. [15] together with cross-method Pearson correlation and the intersection-over-union of the top 20% most salient pixels.

2.5.2 Design Analysis

The dominant architectural pattern in the contemporary PAD literature is the backbone-and-head decomposition, in which a generic feature extractor pretrained on ImageNet is composed with a task-specific classification head consisting of one or more fully connected layers. The pattern is attractive because it permits the reuse of millions of pretrained weights and reduces the data requirement for the downstream task. The present project adheres to this pattern across all three backbones, replacing the original thousand-class ImageNet classifier with a single-logit binary head and applying a global average pooling layer to collapse the spatial dimensions. The transfer-learning regime is configured as full fine-tuning rather than feature-extraction-only, since preliminary experiments indicated that the low-level filter banks learned on natural images do not optimally capture ridge and pore textures.

The insertion point of attention modules is a design decision with significant downstream implications. Woo et al. [10] originally placed CBAM after every residual block, but the present project follows the more conservative pattern of inserting one module after each of the four residual stages, balancing parameter overhead against the granularity of attention. The placement decision interacts with the choice of XAI method, since Grad-CAM++ extracts gradients with respect to the final convolutional stage and therefore benefits from attention modules positioned just upstream of that stage. SHAP DeepExplainer, by contrast, propagates contributions back to the input pixels and is comparatively insensitive to the internal placement of attention. The empirical consequence, observed in the faithfulness experiments described in Chapter 4, is that the three XAI methods exhibit cross-method Pearson correlations below 0.25 across both ResNet50+CBAM and MobileNetV3-Large, indicating that the methods are near-orthogonal and that their union supplies strictly more explanatory information than any single method.

Threshold selection determines the operating point at which the continuous classifier output is binarised into a live or spoof decision. The equal error rate (EER) threshold, defined as the operating point at which the false-acceptance and false-rejection rates coincide, was adopted as the default in the present project, with secondary thresholds tuned for low-BPCER and low-APCER operating regimes. The choice of EER threshold is consistent with the LivDet reporting convention and permits direct comparison against [4], [5], and [6]. From a forensic admissibility perspective, however, the operating threshold is itself a piece of evidence that must be disclosed alongside the prediction, since it materially affects the false-positive and false-negative rates that the Daubert standard requires to be quantified.

Forensic admissibility imposes a further set of design constraints that ordinary biometric systems do not face. Each prediction generated by the deployed Streamlit application is bound to an immutable audit record containing the case identifier, the operator identifier, the cryptographic hash of the input image, the model commit hash, the predicted label, the probability score, the operating threshold, and the timestamps of upload, inference, and explanation generation. The generated PDF report includes all three XAI overlays alongside the deletion-AUC and insertion-AUC values for the prediction, supplying the

trier of fact with both the model's output and a quantitative measure of its explanation faithfulness. This design directly addresses the testability and known-error-rate prongs of the Daubert standard and operationalises the recommendation of Adebayo et al. [18] that saliency methods should be paired with sanity checks before deployment.

2.5.3 Workflow Analysis

The standard PAD training and evaluation workflow follows a stage-wise pipeline of data ingestion, preprocessing, augmentation, model training, validation-based checkpointing, and held-out test evaluation. The present project augments the standard pipeline at three points. First, the combined LivDet 2009-2015 partition is constructed by harmonising the per-edition metadata schemas and resolving sensor naming conflicts, producing a unified corpus of approximately 65,267 images split into 35,427 training images (17,952 live, 17,475 spoof) and 29,840 test images (14,376 live, 15,464 spoof). Second, augmentation is restricted to rotation, horizontal flip, and small affine perturbations, deliberately excluding colour jitter on the grounds that fingerprint sensors produce monochrome or near-monochrome imagery and that aggressive intensity augmentation can mask the very texture cues that distinguish live from spoof. Third, checkpointing is governed by the validation AUC rather than validation loss, since AUC is the metric most directly comparable across LivDet editions.

Cross-sensor and cross-material evaluation protocols extend the standard workflow by partitioning the test data along axes orthogonal to the training distribution. In the cross-sensor protocol, the model is trained on a subset of sensors and evaluated on the remaining sensors, exposing any sensor-specific features that the network may have memorized. In the cross-material protocol, the analogous partition is performed over the spoof material axis, with materials such as Ecoflex or Liquid Ecoflex held out. The combined LivDet 2009-2015 training regime adopted in this project mitigates but does not eliminate the cross-sensor degradation reported by [3] and [5], and the explicit characterization of the residual gap forms part of the empirical contribution.

The XAI gallery generation workflow forms a parallel pipeline that operates on the trained checkpoints rather than on raw images alone. For each test image, the workflow generates Grad-CAM++ heatmaps, SHAP attribution maps, and LIME superpixel

overlays, computes the deletion-AUC and insertion-AUC scores, and serializes the results to a structured archive that the Streamlit application can subsequently index. The gallery workflow is parameterized by the backbone, the layer at which Grad-CAM++ extracts gradients, the number of SHAP background samples, and the number of LIME perturbations, with all parameters logged to ensure reproducibility. The faithfulness metrics of Samek et al. [20] and Petsiuk et al. [15] are aggregated across the test partition to produce model-level and method-level scores that are reported in the practitioner-facing PDF.

The deployment integration pattern envisaged for the system follows the sensor-to-service-to-system topology common in production biometric pipelines. The fingerprint sensor SDK forwards captured imagery to a model service exposing a REST-style inference endpoint, which returns the prediction, the probability score, and the bundle of XAI overlays. The model service is in turn integrated with the existing organisational system, which in the forensic case is a case management database that records the audit trail and triggers the PDF report generation upon examiner sign-off. This three-tier topology cleanly separates acquisition, inference, and case management, and it permits the model service to be upgraded or audited independently of the surrounding infrastructure. The Streamlit application developed in this project realises the inference and reporting tiers and exposes a simulated sensor input to permit end-to-end demonstration without dedicated capture hardware.

2.6 Reflection and Research Gap

The surveyed literature converges on three observations that jointly justify the research gap addressed by this project. First, intra-dataset accuracy on the LivDet benchmarks has approached saturation, with [4] reporting 99.72% on LivDet 2013 and [6] reporting 97.78% with attention-enhanced ResNet50, yet none of these works extends its evaluation to multi-method explainability, and none reports quantitative faithfulness metrics. Second, the cross-sensor and cross-material generalization gap documented by [3] and [5] persists across architectural families and is not adequately mitigated by intra-dataset training, which motivates the combined LivDet 2009-2015 corpus adopted here. Third, the forensic admissibility frameworks of Daubert and Frye, although extensively

discussed in the legal and forensic-science literature, have not been operationalized into a concrete set of design artefacts for fingerprint PAD systems, leaving an implementation gap that this project closes through the audit trail, model commit logging, and signed PDF reports described in Chapter 4.

The absence of multi-XAI comparative studies in fingerprint PAD is particularly striking. Grad-CAM++ [12], SHAP [13], and LIME [14] are routinely applied in medical imaging and natural-image classification, but their joint application to fingerprint PAD has not been systematically reported, and the cross-method Pearson correlations of below 0.25 observed in the present project suggest that the field has been relying on whichever single method is computationally convenient rather than on the union of complementary attributions. The sanity-check recommendations of Adebayo et al. [18] and the evaluation framework of Samek et al. [20] have likewise not been transferred into the PAD literature, and the deletion-insertion AUC protocols of Petsiuk et al. [15] have been applied only sporadically. The present project therefore contributes the first quantitative XAI faithfulness evaluation in the fingerprint PAD domain across three methods and three backbones, providing empirical grounding for the methodological choices that future workers may inherit.

The absence of practitioner-informed evaluation completes the triangulation of the research gap. The PAD literature is overwhelmingly driven by algorithmic metrics computed on held-out partitions, with little engagement with the digital forensic examiners, biometric system administrators, and identity verification analysts who would ultimately consume the explanations produced by such systems. The thirty-two-item, nine-section practitioner survey instrument developed for the present project and scheduled for deployment in August 2026 directly addresses this gap, soliciting structured feedback on perceived trust, examination workflow integration, and admissibility-readiness of the generated reports. Taken together, the three identified gaps multi-XAI comparison, quantitative faithfulness, and practitioner-informed evaluation and that constitute the research space that this thesis occupies.

Chapter 3 builds on this literature foundation by articulating the research methodology in detail. The Design Science Research Methodology of Peffers et al. [16] is adopted as the

overarching strategy, structured into the six canonical stages of problem identification, objective definition, design and development, demonstration, evaluation, and communication. The chapter operationalises the gaps identified above into concrete research questions, hypotheses, and evaluation protocols, and it specifies the experimental design, ethical considerations, and project management approach under which the empirical work was conducted.

Chapter 3 - Methodology

This chapter presents the methodological foundation underpinning the design, development and evaluation of the explainable fingerprint spoof detection artefact. It articulates the philosophical paradigm guiding the research, the reasoning approach adopted, the overarching research strategy, the mechanisms employed for fact collection, and the structured workflow used to operationalise the chosen strategy. The chapter also discusses the project management methodology, the indicative timeline, and the ethical considerations governing the conduct of the work. Each methodological choice is justified in relation to the research questions established in Chapter 1 and the gaps identified in Chapter 2, ensuring traceability between epistemology and execution. The chapter concludes by summarising the methodological position adopted and by signposting the implementation chapter that follows.

3.1 Research Paradigm

A research paradigm constitutes the philosophical worldview that shapes how the investigator perceives reality, knowledge, and the legitimate methods of enquiry. In this study the paradigm adopted is pragmatism, which holds that the value of a research approach is determined by its capacity to produce actionable knowledge for the problem at hand rather than by adherence to a single ontological doctrine. The research problem making fingerprint presentation attack detection both accurate and interpretable for forensic deployment. That is inherently practical: it demands quantitative measurement of model behaviour and, simultaneously, an understanding of how human practitioners perceive and consume model explanations [16]. Pragmatism legitimises the combination of these complementary data forms within a single coherent investigation.

A purely positivist stance was considered and rejected. Positivism privileges objective measurement and assumes that social phenomena, including practitioner trust in algorithmic outputs, can be reduced to measurable variables in a controlled setting. While positivism is well suited to the model-performance dimension of this work, where deletion-area-under-the-curve, insertion-area-under-the-curve, cross-method Pearson correlation, and intersection-over-union at the top 20% threshold provide replicable

numerical evidence [15][20], it is poorly suited to the third research question, which concerns the lived professional perception of explanation usefulness. Conversely, an interpretivist paradigm, which treats reality as socially constructed and privileges qualitative meaning-making, would adequately frame the practitioner survey but would be inappropriate for benchmarking three convolutional architectures against established LivDet error rates.

Pragmatism, by accepting that quantitative and qualitative inferences are complementary rather than rival, reconciles these two epistemic demands within a single research design. It also aligns with the design science orientation discussed in Section 3.3, which prizes artefacts that work in practice over theoretical abstractions. Within this paradigm, truth claims about classifier accuracy are grounded in repeatable experimentation on the LivDet 2009-2015 corpus, whereas truth claims about explanation usefulness are grounded in the situated judgements of forensic and biometric practitioners, gathered through a structured survey instrument. The paradigm therefore provides the philosophical license to combine these evidence streams without violating internal consistency.

3.2 Research Approach

Consistent with a pragmatist paradigm, the research adopts a mixed reasoning approach that integrates deductive and inductive logics. The deductive component operates on the model-performance dimension. Specific hypotheses derived from prior literature for instance, that attention-augmented backbones do not necessarily outperform parameter-efficient backbones on cross-sensor data [3][5], or that gradient-based and perturbation-based explanation methods produce mutually weakly correlated saliency [12][14][20] are operationalized as falsifiable predictions and tested against measured outcomes on the held-out LivDet test partition of 29,840 images. Deductive reasoning is appropriate here because the hypotheses are well-defined, the dependent variables are numerical, and the experimental conditions can be controlled through pinned random seeds, fixed image preprocessing, and a single evaluation harness.

The inductive component operates on the practitioner-perception dimension. Open-ended responses gathered through the thirty-two item survey will be analyzed thematically, with codes emerging from the data rather than imposed in advance. Inductive reasoning is appropriate here because the constructs of interest is trust calibration, perceived forensic admissibility, and explanation utility under courtroom-style scrutiny are not yet sufficiently theorised to permit confirmatory testing. By combining deductive testing of measurable model behaviour with inductive theorising about practitioner reception, the mixed approach produces a more comprehensive answer to the overall research question than either logic alone could supply, and it preserves the methodological coherence required by the pragmatist paradigm.

3.3 Research Strategy

The research strategy selected is the Design Science Research Methodology (DSRM) articulated by Peffers et al. [16]. DSRM is a structured framework for conducting research whose central deliverable is a purposeful IT artefact. In this case a trained ensemble of convolutional classifiers, an explanation pipeline integrating Grad-CAM++ [12], SHAP [13], and LIME [14], and a Streamlit forensic web application with Daubert/Frye aligned PDF reporting. DSRM prescribes six activities. Those are problem identification, definition of objectives, design and development, demonstration, evaluation, and communication. Its explicit emphasis on artefact construction and evaluation aligns closely with the research questions guiding this study.

Three alternative strategies were considered and rejected. Action research was unsuitable because it presupposes embedded participation within an operating organisation and an explicit change-agent role, neither of which is feasible for an undergraduate investigator working from outside the forensic laboratories whose practice the research aims to inform. A pure experimental strategy would have privileged hypothesis testing on model accuracy but would have produced no deployable artefact and could not have engaged the third research question on practitioner perception. A case study strategy would have offered rich contextual understanding of a single forensic unit but would have sacrificed external validity across sensors, materials, and institutional contexts.

DSRM accommodates the dual nature of this work. It permits experimental rigour during the model-evaluation phase, structured demonstration through the Streamlit artefact, and a clearly defined evaluation phase in which both quantitative faithfulness metrics and qualitative practitioner feedback inform iterative refinement. It also makes the deliberate iteration between stages explicit, which is essential in a project where the design of the explanation pipeline was informed by intermediate evaluation outcomes such as the observed cross-method correlation falling below 0.25 on both architectures.

3.4 Fact Collection Mechanisms

Fact collection in this research draws on two distinct yet complementary streams: secondary data for model training and evaluation, and primary data for practitioner perception. Each stream is matched to the research question it serves, ensuring that no evidential burden is placed on a source unsuited to bear it. The secondary stream supplies the volume and standardisation required for credible deep learning benchmarks, while the primary stream supplies the situated professional judgement required to evaluate the practical utility of the explanations produced by the system.

The secondary data comprise the combined LivDet 2009, 2011, 2013, and 2015 corpora, freely available from the LivDet consortium portal. The combined corpus contains approximately 65,267 fingerprint images distributed across nine sensors and twelve spoof materials, with a natural class balance of roughly 51% live to 49% spoof. After deduplication and channel-aware preprocessing, 35,427 images (17,952 live and 17,475 spoof) are used for training and 29,840 images (14,376 live and 15,464 spoof) for testing. The use of secondary data is justified on three grounds: the LivDet competitions are peer-reviewed and have served as the de facto benchmark for fingerprint presentation attack detection since 2009 [5]; results are therefore directly comparable to those reported in prior literature [1][2][6]; and the corpora include sensor and material metadata that permit per-subgroup error reporting, which is critical for the bias analysis discussed in Section 3.8.

The primary data are collected through an anonymous, voluntary practitioner survey of thirty-two items organised in nine sections, distributed to forensic examiners, biometric

researchers, and digital forensic analysts. The survey was constructed because the third research question concerning the perceived usefulness, comprehensibility and admissibility of the explanation outputs cannot be answered from secondary data, however large. The instrument captures Likert-scale judgements on explanation clarity and trust, comparative preferences across Grad-CAM++, SHAP, and LIME outputs, and open-ended responses on perceived forensic admissibility under Daubert and Frye standards. Recruitment is conducted through professional networks and biometric research mailing lists, with deployment planned for August 2026 following ethics approval as described in Section 3.8.5.

3.5 Research Methodology Execution Workflow

To operationalise the DSRM strategy described above, the research is decomposed into a sequence of stages mapped to specific activities and outcomes within the present study. Table 3.1 summarises this mapping. Each row corresponds to a recognisable DSRM activity, the second column explains how the activity is enacted in the context of fingerprint presentation attack detection, and the third column records the concrete outcome produced. The table is intended as a navigational aid rather than a strict linear schedule: as discussed below, several feedback loops connect later stages back to earlier ones.

DSRM Stage	How Addressed in This Research	Outcome
Problem Identification	Survey of recent presentation attack incidents (Kuwait silicone attendance fraud January 2025; wax-spoof court-record falsification) combined with a structured review of the gap between black-box accuracy and forensic admissibility requirements under Daubert and Frye standards.	A precisely scoped problem statement focused on the explainability deficit in CNN-based fingerprint PAD for digital forensics.

Relevance Justification	Triangulation across forensic policy literature, biometric standards, and judicial admissibility doctrine to demonstrate why interpretability is not optional in evidentiary contexts.	Documented justification linking the technical contribution to a recognised operational and legal need.
Comparative Analysis and Gap Justification	Critical review of Slim-ResCNN [5], dual VGG16+ResNet50 [4], attention-augmented ResNet50 [6] and CNN-GAN hybrid [7] approaches against the explainability and cross-sensor generalisation criteria established in Chapter 2.	Identification of the unfilled niche: quantitatively validated, multi-method XAI integrated within a forensic workflow.
Define and Finalise Objectives	Translation of the identified gap into measurable objectives covering classifier accuracy, XAI faithfulness, cross-method orthogonality, and practitioner-perceived usefulness.	Five research objectives mapped one-to-one to evaluation procedures.
Design / Development / Data Management and Handling	Implementation of three backbones (ResNet50V2, ResNet50+CBAM, MobileNetV3-Large) in PyTorch 2.x with ImageNet-pretrained weights, integration of Grad-CAM++, SHAP DeepExplainer (with GradientExplainer fallback) and LIME, and construction of a Streamlit forensic UI with audit trail and PDF reporting.	Functioning artefact reproducible from a pinned-dependency GitHub repository.
Evaluation and Communication	Quantitative evaluation using AUC, ACE, deletion-AUC,	Triangulated evidence base demonstrating that

	insertion-AUC, Pearson correlation, and IoU at top 20% [15][20]; qualitative evaluation through the practitioner survey; communication via this thesis, supervisor review, and intended conference dissemination.	MobileNetV3-Large offers the most favourable accuracy-efficiency-explainability trade-off (AUC 0.9924, ACE 4.60 20%) and that cross-method XAI correlation remains below 0.25.
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Table 3.1: DSRM Execution Workflow

Although Table 3.1 presents the stages in their canonical order, execution involved several deliberate iteration loops between stages. The most significant loop ran from the evaluation stage back to the design and development stage: the observation that cross-method Pearson correlation between Grad-CAM++, SHAP, and LIME remained below 0.25 across both ResNet50V2 and MobileNetV3-Large prompted re-examination of the explanation pipeline configuration, addition of the deletion-insertion protocols proposed by Petsiuk et al. [15], and inclusion of intersection-over-union at top 20% as an additional alignment metric. A second loop ran from design back to relevance justification: the practical observation that MobileNetV3-Large delivered approximately four-fold faster XAI inference on CPU and used 82% fewer parameters than ResNet50 reinforced the forensic relevance argument for parameter-efficient backbones.

A further loop linked the demonstration stage to the fact-collection mechanisms. Early prototypes of the Streamlit forensic UI exposed ambiguities in how examiners would interpret the heatmap legends, which in turn shaped the wording of Likert items in the practitioner survey. This kind of bidirectional traffic between artefact construction and instrument design is one of the practical strengths of DSRM as a strategy for socio-technical research, and it could not have been achieved within a strictly linear waterfall workflow.

3.6 Project Management Methodology

The project management methodology adopted is Kanban, configured for a single-developer undergraduate research context. Kanban is a continuous-flow approach in which work items are visualised on a board, work-in-progress limits are enforced per

stage, and the cadence of delivery is governed by the natural completion rate of each item rather than by fixed time boxes. This characteristic is well suited to a research project in which the scope of subsequent tasks is frequently revealed only through the outcomes of preceding investigations. For example, the decision to add the CBAM-augmented backbone arose from observations during the initial ResNet50V2 training cycle, and the decision to include intersection-over-union at top 20% arose from inspection of early SHAP outputs.

Two alternatives were considered and rejected. Scrum was unsuitable because its ceremonies are sprint planning, daily stand-ups, sprint reviews and retrospectives. PRINCE2 methodology was unsuitable because its emphasis on stage-gated approvals, formal business cases and project boards is over-formal for an individual research dissertation and would have consumed disproportionate effort relative to the technical work. Kanban delivers the necessary visibility of work-in-progress, supports flexible re-prioritisation when investigation surfaces new tasks, and aligns naturally with the iterative loops between DSRM stages described in Section 3.5.

3.7 Project Timeline

The project schedule spans the period from initial topic ratification through to final submission in November 2026 and is depicted in Figure 3.1. The schedule is divided into discrete phases covering literature review, dataset acquisition and preprocessing, backbone training, XAI integration, faithfulness evaluation, Streamlit artefact development, ethics approval, survey deployment, thematic analysis, and thesis writing. Milestones are aligned with academic submission dates, including the present Interim Submission 01 covering Chapters 1 to 3, and the final viva examination.

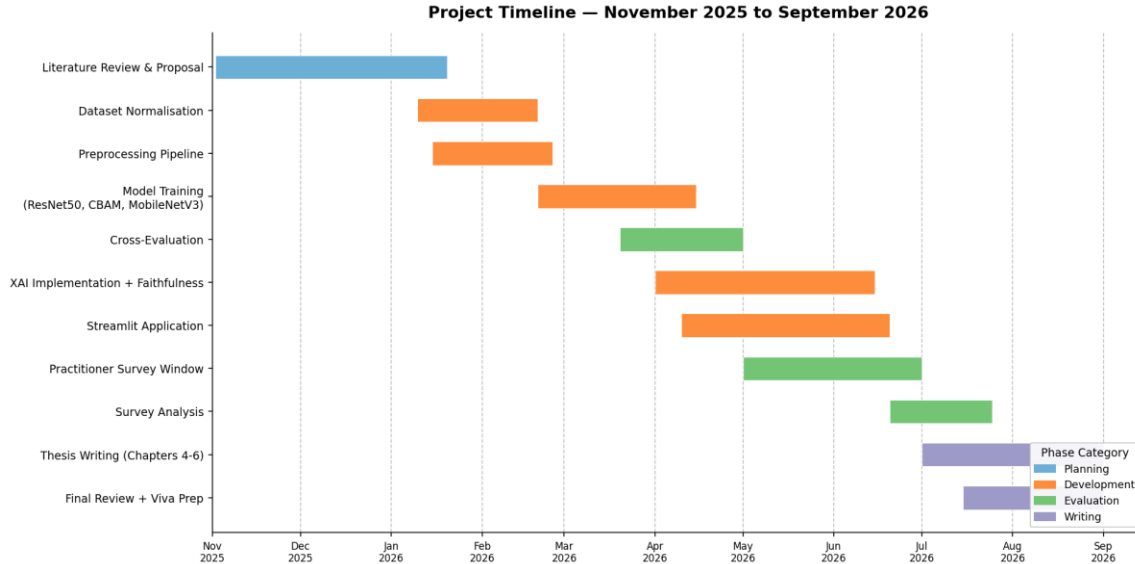


Figure 3.1: Project Timeline (Gantt Chart)

Three phases lie on the critical path and therefore receive heightened risk attention. The first is model training, since cumulative training cost on the Colab T4 GPU constrains the number of architectural variants that can be explored within the available compute budget; this risk was mitigated by adopting ImageNet-pretrained weights and by using mixed-precision training. The second is XAI faithfulness evaluation, which requires repeated forward passes per image and is computationally heavier than classification inference; this risk was mitigated by the selection of MobileNetV3-Large as a parameter-efficient backbone and by batching the deletion and insertion protocols. The third is the practitioner survey, whose recruitment and response window directly governs the timing of the thematic analysis and consequently the discussion chapter; this risk is mitigated by early ethics approval, multi-channel recruitment, and a deployment window beginning in August 2026 to leave four months of buffer before final submission.

3.8 Ethical Considerations

Ethical considerations are addressed at five levels, ranging from the handling of biometric data through algorithmic fairness, scientific reproducibility, the dual-use risk inherent in biometric anti-spoofing research, and institutional governance. Each level is treated in a dedicated subsection below.

3.8.1 Data Privacy

The fingerprint imagery used for model training and evaluation is drawn exclusively from the publicly available LivDet 2009, 2011, 2013, and 2015 corpora, which were released by their originating consortia under terms permitting academic research. No identifying metadata accompany the images, and no attempt is made to re-identify donors. No new biometric samples are collected from human participants in this work; all biometric data are secondary. Consequently the research does not engage the categories of special-category personal data that would require an additional data protection impact assessment. The Streamlit artefact, when used for demonstration, processes only researcher-supplied images held locally on the development workstation and does not transmit imagery to third-party services.

3.8.2 Algorithmic Bias and Fairness

Reporting only aggregate accuracy metrics on a multi-sensor multi-material corpus can mask systematic disparities in performance. Cross-sensor generalization is a known weakness of fingerprint presentation attack detectors [3][5], and material-specific failure modes can render an otherwise high-accuracy model unsuitable for deployment against particular threat profiles. To mitigate this risk, evaluation in this work is reported not only in aggregate but also broken down per sensor (Biometrika 400B, CrossMatch, DigitalPersona U.are.U 4500, GreenBit DactyScan, HiScan, Italdat ET10, Sagem MorphoSmart, Swipe and Digital 2009) and per material (Silicone, Gelatin, Latex, Wood Glue, Play-Doh, Ecoflex, Body Double, Modasil, OOMOO, RTV, Silgum and Liquid Ecoflex). This per-subgroup reporting permits readers, peer reviewers and forensic deployers to assess whether the system is reliable on the specific sensor-material combinations relevant to their operating context.

3.8.3 Reproducibility

Reproducibility is treated as an ethical obligation because forensic claims must be independently verifiable [18]. To this end, the entire pipeline is implemented in Python 3.13 with pinned dependencies in a requirements specification, including PyTorch 2.x, torchvision, pytorch-grad-cam, shap and lime, and is released through a public GitHub

repository. Random seeds for PyTorch, NumPy and Python's standard library are fixed prior to each experimental run, and deterministic algorithm flags are enabled where the underlying CUDA kernels permit. Training history, including loss and accuracy per epoch, is logged to disk; checkpoints are saved for the best validation accuracy and the final epoch. The Streamlit application reads these checkpoints directly so that any reader with access to the repository can reproduce the published metrics within numerical tolerance.

3.8.4 Dual-Use Considerations

Anti-spoofing research is dual-use in nature: detailed knowledge of which materials defeat which sensors can inform both defensive countermeasures and offensive spoof construction. The present work mitigates this risk in two ways. First, the scope is strictly defensive: no generative model is trained for spoof image synthesis, and no procedural recipe for fabricating spoofs is documented in this thesis. Second, all spoofing materials discussed are already documented in the public LivDet competition reports, so no new offensive capability is introduced. The XAI outputs, while highlighting regions relevant to the live-versus-spoof decision, do not yield material-level fabrication guidance, and any residual risk is judged proportionate to the defensive benefit of equipping forensic practitioners with interpretable detectors.

3.8.5 Institutional Ethics Approval

The practitioner survey component is conducted under the oversight of the NSBM Green University Faculty of Computing Ethics Committee. The instrument is anonymous and voluntary; no personally identifying information is solicited. Respondents are informed at the point of consent of the purpose of the study, the indicative completion time, and their right to withdraw. A seven-day withdrawal window is provided, supported by a six-character self-issued participation code that allows a respondent to request deletion of their responses without disclosing their identity. Survey data are retained for five years from the date of submission, after which they are permanently deleted in accordance with the committee's data retention policy. The supervisor, Ms. Hirushi Dilpriya, reviews all changes to the instrument prior to deployment.

3.9 Chapter Summary

This chapter has established the methodological foundation of the research. A pragmatist paradigm was adopted to legitimise the integration of quantitative model evaluation and qualitative practitioner perception. A mixed reasoning approach, combining deductive testing of model-performance hypotheses with inductive thematic analysis of practitioner responses, was justified as the natural corollary of that paradigm. The Design Science Research Methodology of Peffers et al. [16] was selected as the research strategy because the central deliverable is a purposeful IT artefact. Fact collection draws on the public LivDet 2009-2015 corpora for secondary data and an anonymous thirty-two item practitioner survey for primary data. The execution workflow was operationalised through the six DSRM stages summarised in Table 3.1, with explicit iteration loops between evaluation and design. Project management uses Kanban, suited to single-developer continuous-flow research, and the indicative schedule is depicted in Figure 3.1. Ethical considerations were addressed across data privacy, fairness, reproducibility, dual-use risk, and institutional approval.

With the methodological position established, Chapter 4 turns to the design and implementation of the artefact itself, detailing the data preprocessing pipeline, the three convolutional backbones, the integration of Grad-CAM++ [12], SHAP [13] and LIME [14], and the architecture of the Streamlit forensic web application. Chapter 5 will then report the results of the quantitative evaluation, and Chapter 6 will analyse the practitioner survey responses, before Chapter 7 consolidates the contributions and discusses limitations and future work.

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